

A forecast reconciliation approach to cause-of-death mortality modeling

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ARTICLE INFO

Article history:

Received September 2018

Received in revised form January 2019

Accepted 25 February 2019

Available online 5 March 2019

Keywords:

Cause-of-death

Mortality modeling

Forecast reconciliation

Hierarchical time series

Clustering

Cause-elimination

ABSTRACT

Life expectancy has been increasing sharply around the globe since the second half of the 20th century. Mortality modeling and forecasting have therefore attracted increasing attention from various areas, such as the public pension systems, commercial insurance sectors, as well as actuarial, demographic and epidemiological research. Compared to the aggregate mortality experience, cause-specific mortality rates contain more detailed information, and can help us better understand the ongoing mortality improvements. However, when conducting cause-of-death mortality modeling, it is important to ensure coherence in the forecasts. That is, the forecasts of cause-specific mortality rates should add up to the forecasts of the aggregate mortality rates. In this paper, we propose a novel forecast reconciliation approach to achieve this goal. We use the age-specific mortality experience in the U.S. during 1970–2015 as a case study. Seven major causes of death are considered in this paper. By incorporating both the disaggregate cause-specific data and the aggregate total-level data, we achieve better forecasting results at both levels and coherence across forecasts. Moreover, we perform a cluster analysis on the cause-specific mortality data. It is shown that combining mortality experience from causes with similar mortality patterns can provide additional useful information, and thus further improve forecast accuracy. Finally, based on the proposed reconciliation approach, we conduct a scenario-based analysis to project future mortality rates under the assumption of certain causes being eliminated.

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1. Introduction

Increases in human lifespan around the globe have wide reaching implications for health care, pension systems and the insurance industry. This has urged the need to better understand underlying patterns in mortality improvements and exploit these patterns to accurately project mortality rates into the future (for a review of the extensive literature on mortality modeling and forecasting see [Booth and Tickle, 2008](#)). Modeling mortality has not been restricted to all-cause mortality rates; the study of cause-specific mortality rates has been fruitful both in identifying the key drivers of future mortality improvement, and in conducting scenario-based demographic forecasting ([Manton et al., 1976](#); [Keyfitz, 1977](#); [Manton et al., 1986](#); [McNown and Rogers, 1992](#); [Tabeau et al., 1999](#); [Caselli et al., 2006](#); [Dimitrova et al., 2013](#);

[Arnold and Sherris, 2013, 2015](#); [Alai et al., 2015, 2018](#); [Giordano et al., 2018](#)). Forecasting all-cause mortality rates and individual cause-specific mortality rates are both important empirical questions in their own right but the best forecasting approach may differ for each individual series. This results in forecasts that are not coherent, in the sense that forecasts of disaggregate cause-of-death mortality rates do not add up to the forecast of the aggregate. Taken together, all-cause and cause-specific mortality rates are well suited to recent advances in forecast reconciliation methodology (for an overview see [Hyndman and Athanasopoulos, 2018](#), Chapter 10). This methodology not only ensures coherence but also improves forecast accuracy. Investigating whether the application of forecast reconciliation methods to cause-of-death mortality data improves forecast accuracy is thus the primary motivation of our study.

Analyses of cause-of-death mortality often involve joint risk modeling, for example multinomial logistic regression models (see e.g. [Michael et al., 1999](#); [Alai et al., 2015, 2018](#)), multivariate time series models (see e.g. [Arnold and Sherris, 2013, 2015, 2016](#))

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and copula models (see e.g. Frees and Valdez, 1998; Escarela and Carriere, 2003; Kaishev et al., 2007; Dimitrova et al., 2013; Li and Lu, 2018). Whenever these studies have forecast all-cause mortality rates, they have done so in a bottom-up fashion, i.e. forecasts of all cause-of-death mortality rates are produced jointly and then aggregated to produce a forecast for the total. Consequently, a shortcoming that all these approaches share – at least from the forecasting perspective – is a failure to exploit information obtained from directly modeling the time series of all-cause mortality rates. Wilmoth (1995) found that mortality projections made in a bottom-up fashion are always more pessimistic relative to those formed using the aggregate series alone, suggesting that forecast improvement can be achieved by combining forecasts based on different datasets.

Time series data that follow an aggregation structure, such as cause-of-death mortality rates, are also known as hierarchical time series, and are common across a number of disciplines. In this context forecast reconciliation has emerged as a state-of-the-art methodology that achieves two main goals. First, reconciliation ensures that forecasts are coherent, which can be crucial for aligned decision making. Second, reconciliation improves forecast quality by combining information from time series at both aggregate and disaggregate levels. A common finding is that forecast reconciliation methods consistently produce better forecasts than a bottom-up approach even when forecasts from the bottom level are produced using a joint model (see, e.g. Athanasopoulos et al., 2009; Capistrán et al., 2010; Borges et al., 2013; Syntetos et al., 2016; Wickramasuriya et al., 2018). This arises since even the most sophisticated dependence models are prone to misspecification, parameter uncertainty and the curse of dimensionality. Thus, simply adding up bottom-level forecasts may amplify these problems in producing higher-level forecasts. While information can be lost in the data aggregation process, much of the noise is also smoothed out. By combining forecasts at all levels, forecast reconciliation methods get the best of both worlds.

We conduct an empirical case study based on the U.S. male population mortality data during 1970–2015.¹ The seven leading causes of death considered in our study are *cancer*, *diabetes*, *external*, *influenza*, *mental*, *nephritis*, and *vascular*. The minimum trace (MinT) method of Wickramasuriya et al. (2018) is adopted as a reconciliation method due to its optimal theoretical properties relative to the bottom-up approach and other reconciliation methods. An additional major advantage of MinT is that dependence in forecast errors is taken into account in the reconciliation process. This allows us to obtain first-stage unreconciled forecasts by fitting a univariate Lee–Carter model (Lee and Carter, 1992) to each cause and to the total, thus circumventing the need for heavily parameterized and possibly misspecified joint models subject to the curse of dimensionality.

Our contribution to the literature is three-fold. First, we show that more accurate and coherent forecasts can be achieved for both cause-specific mortality rates and total mortality rates using MinT reconciliation. While the earlier reconciliation method of Hyndman et al. (2011) is applied to Japanese regional mortality rates by Shang and Hyndman (2017) and Shang and Haberman (2017), to the best of our knowledge, our study is the first to apply state-of-the-art forecast reconciliation methods to cause-of-death mortality modeling. Second, we adapt the forecast reconciliation methodology to quantify the effect of cause-elimination on future mortality projections under hypothetical scenarios, which is an important and interesting topic in cause-specific mortality modeling. Third, we show that forecasts can be further improved by

using cluster analysis to group together bottom level series and adding these clusters to the hierarchical time series. The cluster analysis reveals the causes of death with similar mortality profiles and improves our understanding of these data. Furthermore, the use of cluster analysis for forecast reconciliation is novel since the hierarchical structure is usually assumed as given. This methodological innovation makes a valuable contribution to the literature on forecast reconciliation that could be applied in other empirical settings.

The rest of the paper is organized as follows. Section 2 describes the data, and conducts a preliminary cluster analysis of the causes of death. Section 3 elaborates the forecast reconciliation approach. Section 4 is devoted to the empirical implementation of the method on the U.S. male mortality data. Conclusions are drawn in Section 5.

2. Data

2.1. Notation

We start the section by defining the notation which will be used throughout this paper. Suppose that we observe J mutually exclusive causes-of-death, for X different ages and at T calendar years. For $j \in (1, J)$, $x \in (1, N)$ and $t \in (1, T)$, we define the bottom-level mortality series as

$$m_{j,t}(x) = \frac{D_{j,t}(x)}{E_t(x)}, \quad (1)$$

where $D_{j,t}(x)$ is the number of deaths at age x and year t due to cause j , and $E_t(x)$ is the central exposed-to-risk which represents the average population aged x in year t .

These different causes can be combined into a number of clusters, based on their age and time patterns. The cluster-specific mortality rate is defined by

$$m_t^c(x) = \frac{D_t^c(x)}{E_t(x)}, \quad (2)$$

where $D_t^c(x)$ is the number of deaths at age x and time t due to cluster c , and we have

$$D_t^c(x) = \sum_{j \in J_c} D_{j,t}(x), \quad (3)$$

where J_c denotes a subset of causes of death that belongs to cluster c .

Finally, the top-level, all-cause mortality rate for age x and year t is given by

$$m_t(x) = \frac{D_t(x)}{E_t(x)}, \quad (4)$$

where $D_t(x)$ is the corresponding total number of deaths.

2.2. U.S. male mortality data

The empirical study in this paper concerns the U.S. male mortality. In particular, we consider seven major causes of death including *cancer* (i.e. malignant neoplasms), *diabetes*, *external*, *influenza* (including pneumonia), *mental* (including nervous system disorders), *nephritis*, and *vascular* (i.e. major cardiovascular diseases), plus an 8th cause covering all remaining causes of death.

The mortality data come from two sources. First, we obtain age-specific population exposures from the Human Mortality Database (HMD).² Second, we compute the annual, age-by-age cause-specific death counts between age 1 and 85 from the Center

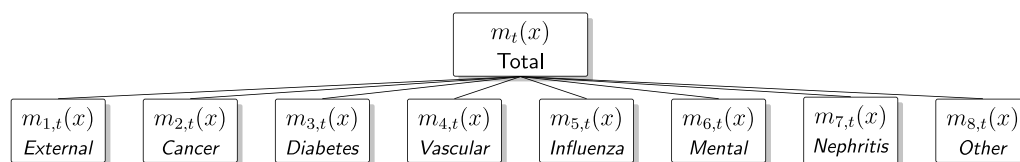
¹ We have also considered female mortality rates from the U.S. for the same period, and find the results in-line with conclusions and findings in the male case. These results are available upon request.

² The HMD mortality database can be found at: www.mortality.org.

Table 1

Codification and tape location of the seven major causes of death.

Cause of death	1970–1978			1979–1998			1999–2015		
	Coding system	NCHS recode	Location	Coding system	NCHS recode	Location	Coding system	NCHS recode	Location
Cancer	ICD8	UCOD_34	050–110	ICD9	UCOD_34	50–110	ICD10	UCOD_39	4–15
Diabetes	ICD8	UCOD_34	120	ICD9	UCOD_34	120	ICD10	UCOD_39	016
External	ICD8	UCOD_34	330–370	ICD9	UCOD_34	330–370	ICD10	UCOD_39	038–042
Influenza	ICD8	UCOD_34	230	ICD9	UCOD_34	230	ICD10	UCOD_39	027
Mental	ICD8	UCOD_281	141–159	ICD9	UCOD_282	141–159	ICD10	UCOD_358	174–194
Nephritis	ICD8	UCOD_34	270	ICD9	UCOD_34	270	ICD10	UCOD_39	031
Vascular	ICD8	UCOD_34	150–220	ICD9	UCOD_34	150–220	ICD10	UCOD_39	018–026

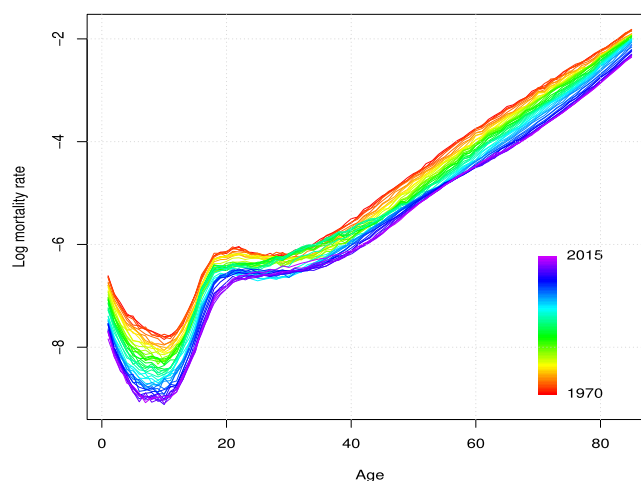
**Fig. 1.** 2-level hierarchical tree of mortality rates.

for Disease Control and Prevention (CDC)³ for the period 1970–2015. While longer historical data is also available as the CDC keeps death records in the U.S. since 1959, we have decided not to include the earlier records in the analysis due to data quality concerns. Indeed, the chief of National Center for Health Statistics (NCHS)⁴ mortality branch has declared that “while the 1959–1967 files are generally OK, they have not been rigorously verified”. Moreover, the coding system used after 1970 is more consistent than earlier records.⁵

The CDC mortality database provides detailed individual-level information, including the date of birth, gender, age at death and the cause-of-death for each deceased. Compared to the alternative cause-specific datasets, e.g., the one published by the World Health Organization, a major advantage of the CDC dataset is that the age of death is not rounded to multiples of 5. Hence, the CDC dataset allows us to construct more accurate age-cause specific mortality rates. In the CDC dataset, the codification of cause-of-death is based on the International Classification of Diseases (ICD), leading to around 1000 different causes. For publication purpose, the NCHS has also recoded the ICD code into a much smaller number of groups.⁶ During our observation window 1970–2015, the ICD coding system has changed twice, from ICD8 to ICD9, then to ICD10. Therefore, the NCHS codifications have also been subject to some amendments.

To better illustrate these changes and the cause-of-death classifications adopted in this paper, in Table 1 we provide a summary of the codifications for the seven major causes of death, as well as the corresponding tape location. After constructing the cause-specific mortality dataset based on Table 1, we group the number of deaths from the remaining causes in the CDC dataset and denote them as *other*. Naturally, the structure of our data can be presented in a hierarchy setting, as shown in Fig. 1.

Based on the collected data, we plot the logarithm of total mortality rates for the U.S. male population during 1970–2015 in Fig. 2. The rainbow plot shows an overall decreasing trend

**Fig. 2.** Rainbow plot for total mortality.

in total mortality rates, although the rate of decline seems to vary across different age groups. On the age dimension, we can observe high infant mortality rates, an “accident hump” around age 20 and steady increases in mortality rates in middle-to-old age groups. The total mortality rate series are fairly smooth and illustrate clear patterns in time and age dimensions.

To gain more insights into the mortality experience, in Fig. 3 we produce similar rainbow plots for the eight causes (including *other*) considered in this paper. We can see that these causes differ substantially in terms of the mortality pattern in the age dimension. For example, *external* shows the most apparent pattern of “accident hump” around the early 20s, whereas for causes such as *diabetes*, *nephritis* and *vascular*, we can hardly observe any sudden increase in mortality curve around that age range. Not surprisingly, different causes of death also show various levels of mortality improvement over time. For example, mortality due to causes such as *cancer*, *external*, *influenza*, and *vascular* has declined over the sample period. On the contrary, those due to *diabetes*, *mental* and *nephritis* seem to have experienced an increase during recent years, in particular for older age groups. Obviously, cause-specific mortality rates reveal more information on both age and time dimensions of the mortality experience. However, it can also be seen that these cause-specific rates are generally noisier than the total mortality rates.

³ The CDC mortality database can be found at: <http://www.nber.org/data/vital-statistics-mortality-data-multiple-cause-of-death.html>.

⁴ The NCHS is a division of the CDC responsible for the collection and dissemination of the Nation's vital and health statistics.

⁵ See more details at: <http://www.nber.org/data/vital-statistics-mortality-data-multiple-cause-of-death.html>.

⁶ The NCHS provides an annual report documenting how these new codifications are constructed from the ICD code. See for instance <http://www.nber.org/mortality/2015/mort2015.pdf> for the explicative notice of year 2015.

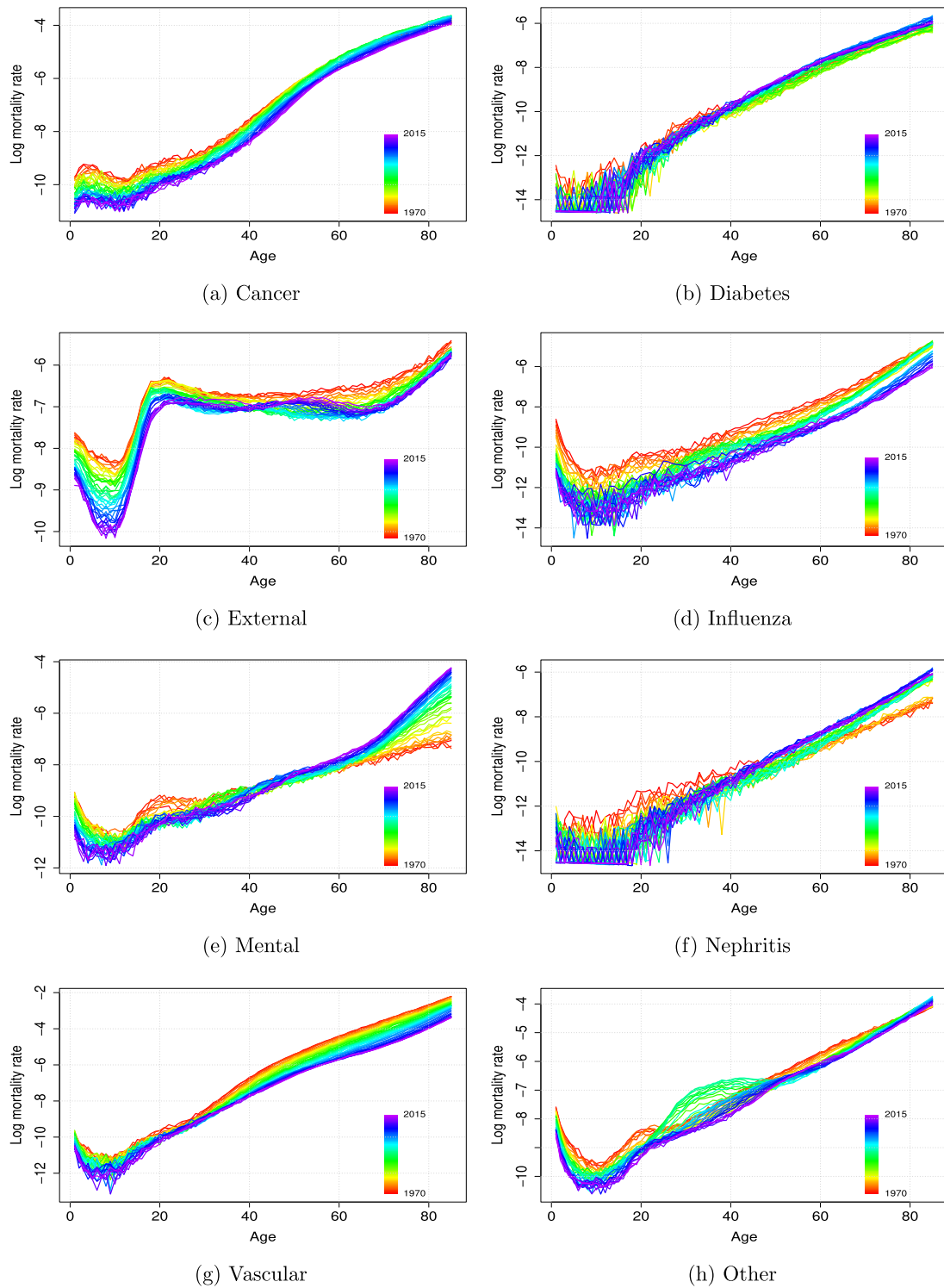


Fig. 3. Rainbow plots for eight causes of death.

2.3. Cluster analysis

Besides visual inspections, in order to statistically determine the degree of similarity between different causes, we also conduct a cluster analysis on the seven major causes of death in this section.

To this end, let us first define a distance measure based on the age–time profile of each cause of death. Specifically, let $\mathbf{M}_j$ be the $T \times N$ matrix of mortality rates due to cause j . The element on

row t and column x of $\mathbf{M}_j$ is given by $m_{j,t}(x)$. Since we intend to cluster the causes based on the shape of their age–time mortality profile rather than the scale, we define a normalized matrix $\tilde{\mathbf{M}}_j = (1/s)\mathbf{M}_j$. Our choice of s is the standard deviation of all elements in $\mathbf{M}_j$, and we have checked that using the interquartile range gives similar results. The distance between cause j and cause k is then given by

$$\delta_{j,k} = \|\tilde{\mathbf{M}}_k - \tilde{\mathbf{M}}_j\|, \quad (5)$$

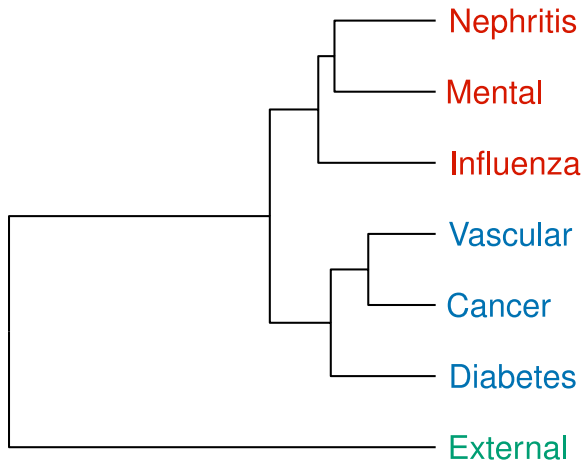


Fig. 4. Dendrogram for agglomerative clustering using Ward's criterion.

where $\|A\|$ denotes a matrix norm of A . We present results for the Frobenius norm defined as $\sqrt{\text{tr}(A'A)}$, where tr is the matrix trace⁷ (Graybill, 1983).

Then we conduct an agglomerative clustering using Ward's criterion (Ward, 1963; Murtagh and Legendre, 2014), which leads to the dendrogram in Fig. 4. Since the category *other* is already a combination of causes it was excluded from the cluster analysis. The remaining seven causes can be grouped into three clusters. The first cluster contains *nephritis*, *mental* and *influenza*. This echoes the rainbow plots in Fig. 3, from which we can see that *nephritis* and *mental* share the common pattern of worsening mortality over time among the elderly. The second cluster contains *vascular*, *cancer* and *diabetes*; the rainbow plots show that these causes have flatter mortality rates over the younger age range. Finally, *external* forms its own cluster due to its idiosyncratic age–time profile, in particular there is a very sharp increase towards early 20s. Based on this cluster analysis, we can reconstruct the hierarchical tree in Fig. 1 into 3 levels. The new hierarchical tree is shown in Fig. 5.

3. A forecast reconciliation approach

Forecast reconciliation is a technique ensuring the coherence of forecasts across all levels in a hierarchy (Hyndman et al., 2014). In the literature, there have been many studies on forecasting hierarchical series (see e.g. Shlifer and Wolff, 1979; Schwarzkopf et al., 1988; Weale, 1988; Gross and Sohl, 1990; Dangerfield and Morris, 1992; Kahn, 1998; Zellner and Tobias, 2000; Athanasopoulos et al., 2009; Hyndman et al., 2011; Wickramasuriya et al., 2018). Traditionally, the “bottom-up” method is the most commonly used technique to reconcile hierarchical time series. This approach involves two steps: first, forecasts are generated for each time series at the bottom level; and second, these forecasts are summed up to form forecasts at higher levels. It is worth noting that the bottom-up method does not take into account all available data in the hierarchy during the forecasting process, as only the most disaggregate time series are used.⁸ In fact, as mentioned earlier, it has been shown that forecast accuracy can be improved by utilizing information at all levels in many applications of hierarchical forecasting.

In this paper, we focus on the bottom-up method, and the trace minimization (MinT) method proposed by Wickramasuriya et al. (2018). For each cause-specific, cluster-specific as well as all-cause data series, we can always forecast the mortality rates based on their own history, using independent models. These forecasts are referred to as “base forecasts” in this study. Ideally, we would like to see the base forecasts to add up in a manner that is consistent with the underlying structure of the hierarchy. However, in reality this is very unlikely to happen. Therefore, we need to reconcile the base forecasts so that they fulfill certain aggregation constraints. For example, in the hierarchical structure shown in Fig. 5, we have the following aggregation constraints:

$$\sum_{j \in \mathcal{C}_c} m_{j,t}(x) = m_t^c(x), \quad \forall c \in (1, \mathbb{C}), \quad (6)$$

$$\sum_{j=1}^J m_{j,t}(x) = m_t(x), \quad (7)$$

for all values of x and t , where $\mathbb{C}$ is the number of clusters with at least two members.

To express these constraints in a matrix form, let

- $\mathbf{y}_t(x) = (m_t(x), m_t^1(x), \dots, m_t^{\mathbb{C}}(x), m_{1,t}(x), \dots, m_{J,t}(x))'$ be a $(\mathbb{C} + J + 1)$ vector containing the observed mortality rates at all levels for age x and time t .
- $\mathbf{b}_t(x) = (m_{1,t}(x), \dots, m_{J,t}(x))'$ be a vector of length J containing the most disaggregated (bottom-level) observed mortality rates for age x and time t .

We can then link these two vectors by

$$\mathbf{y}_t(x) = \mathbf{S} \mathbf{b}_t(x), \quad (8)$$

where $\mathbf{S}$ is a “summing matrix” of dimension $(\mathbb{C} + J + 1) \times J$ which aggregates the most disaggregated series to construct higher level series. As an example, for the hierarchical structure shown in Fig. 5, $\mathbf{S}$ is of dimension 11×8 and is given by

$$\mathbf{S} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ & & & & & & & \mathbf{I}_8 \end{pmatrix}, \quad (9)$$

where $\mathbf{I}_8$ denotes an 8×8 identity matrix. The aggregation constraints (6) and (7) are therefore met by the construction of the $\mathbf{S}$ matrix.

To ensure that base forecasts at each level also add up in the same manner as the observations, adjustments need to be made via forecast reconciliation. According to Wickramasuriya et al. (2018), we can express all linear reconciliation methods as

$$\hat{\mathbf{y}}_{T+h}(x) = \mathbf{S} \mathbf{P} \hat{\mathbf{y}}_{T+h}(x), \quad (10)$$

where $\hat{\mathbf{y}}_{T+h}(x)$ denotes a $(\mathbb{C} + J + 1)$ -vector of h -step-ahead base forecasts of all series in the hierarchy, $\hat{\mathbf{y}}_{T+h}(x)$ denotes a vector containing the corresponding reconciled forecasts of the same length, and $\mathbf{P}$ is the reconciliation matrix of dimension $J \times (\mathbb{C} + J + 1)$. The basic idea behind Eq. (10) is to find a matrix $\mathbf{P}$ which maps a set of base forecasts to a set of reconciled bottom-level forecasts. The selection of $\mathbf{P}$ is not unique.

3.1. The bottom-up method

As mentioned earlier, the bottom-up method is one of the most commonly adopted forecast reconciliation approaches. It simply aggregates bottom-level base forecasts to obtain forecasts at higher levels (Dangerfield and Morris, 1992; Zellner and Tobias, 2000). In doing so, coherence in forecasts throughout the

⁷ We have also implemented the spectral norm, i.e. the square root of the first eigenvalue of $A'A$, and found the same clustering results.

⁸ For more information on the bottom-up method, see Hyndman and Athanasopoulos (2018), Chapter 10, Section 3.

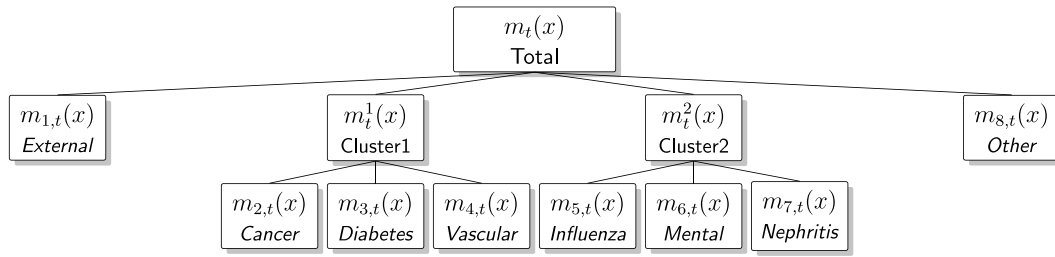


Fig. 5. 3-level hierarchical tree of mortality rates.

hierarchy is guaranteed. To illustrate the bottom-up method in the format of Eq. (10), we have $\mathbf{P}$ chosen as

$$\mathbf{P} = (\mathbf{0}_{J \times (C+1)}, \mathbf{I}_J), \quad (11)$$

where $\mathbf{0}_{J \times (C+1)}$ is a zero matrix of dimension $J \times (C+1)$.

This simply implies that

$$\tilde{\mathbf{y}}_{T+h}(x) = \mathbf{S}\hat{\mathbf{b}}_{T+h}(x), \quad (12)$$

where $\hat{\mathbf{b}}_{T+h}(x)$ is a vector of length J that contains the h -step-ahead base forecasts of the most disaggregate series. Therefore, the top-level h -step-ahead forecasts are given by

$$\tilde{m}_{T+h}(x) = \sum_{j=1}^J \hat{m}_{j,T+h}(x). \quad (13)$$

Besides its simplicity, another advantage of the bottom-up method is that there is no loss of information on the bottom level. However, at the same time, as bottom-level series are generally more noisy, they can be quite challenging to model and thus to forecast (Athanasopoulos et al., 2009; Hyndman et al., 2014). In addition, as the forecasts at higher levels are obtained purely from bottom-level base forecasts, the base forecasts on higher levels are completely ignored in the forecasting process. Finally, and most importantly, the bottom-up method overlooks the variance–covariance structure of the forecasting errors from the disaggregate series.

3.2. The trace minimization method

To make the best use of data and improve forecast accuracy, Hyndman et al. (2011) proposed an optimal combination approach to hierarchical time series forecast reconciliation. The idea behind this method is to utilize all available information and combine the base forecasts at all levels to produce coherent forecasts. Based on this method, the dependency among the time series at different levels is taken into account in the reconciliation process, even if the base forecasts are obtained independently. As mentioned earlier, the optimal combination method has demonstrated superior forecasting performance over the bottom-up method across various disciplines.

Define $\beta_{T+h}(x) = \mathbb{E}[\mathbf{b}_{T+h}(x)|\mathbf{y}_1(x), \mathbf{y}_2(x), \dots, \mathbf{y}_T(x)]$ which is the unknown true mean of bottom-level base forecasts. According to Hyndman et al. (2011), the h -step-ahead base forecasts can be expressed as

$$\hat{\mathbf{y}}_{T+h}(x) = \mathbf{S}\beta_{T+h}(x) + \epsilon_{T+h}(x), \quad (14)$$

where $\epsilon_{T+h}(x)$ has zero mean and a variance–covariance matrix $\Sigma_h(x)$. Therefore, we can obtain the generalized linear squares (GLS) estimator of $\beta_{T+h}(x)$ as

$$\hat{\beta}_{T+h}(x) = (\mathbf{S}'\Sigma_h^{\dagger}(x)\mathbf{S})^{-1}\mathbf{S}'\Sigma_h^{\dagger}(x)\hat{\mathbf{y}}_{T+h}(x), \quad (15)$$

where $\Sigma_h^{\dagger}(x)$ denotes the Moore–Penrose generalized inverse of $\Sigma_h(x)$.

The reconciled forecasts are then produced as

$$\tilde{\mathbf{y}}_{T+h}(x) = \mathbf{S}\hat{\beta}_{T+h}(x) = \mathbf{S}(\mathbf{S}'\Sigma_h^{\dagger}(x)\mathbf{S})^{-1}\mathbf{S}'\Sigma_h^{\dagger}(x)\hat{\mathbf{y}}_{T+h}(x). \quad (16)$$

Therefore, by the optimal combination forecast reconciliation approach, the projection matrix $\mathbf{P}$ is selected as

$$\mathbf{P} = (\mathbf{S}'\Sigma_h^{\dagger}(x)\mathbf{S})^{-1}\mathbf{S}'\Sigma_h^{\dagger}(x). \quad (17)$$

Hyndman et al. (2011) have shown that given the condition of base forecasts being unbiased, the reconciled forecasts from the optimal combination method will also be unbiased. Thus, the question boils down to identifying the variance–covariance matrix $\Sigma_h(x)$. In practice, however, $\Sigma_h(x)$ is generally unknown, and not identifiable (Wickramasuriya et al., 2018). Wickramasuriya et al. (2018) provide a method to approximate $\Sigma_h(x)$, and thus to obtain a closed-form estimation of the matrix $\mathbf{P}$.

Formally, define the h -step-ahead in-sample error of base forecasts as

$$\hat{\mathbf{e}}_{t+h}(x) = \mathbf{y}_{t+h}(x) - \hat{\mathbf{y}}_{t+h}(x), \quad (18)$$

and the corresponding error of the reconciled forecasts as

$$\tilde{\mathbf{e}}_{t+h}(x) = \mathbf{y}_{t+h}(x) - \tilde{\mathbf{y}}_{t+h}(x). \quad (19)$$

It can be shown that⁹

$$\tilde{\mathbf{e}}_{t+h}(x) = \mathbf{S}\mathbf{P}\hat{\mathbf{e}}_{t+h}(x). \quad (20)$$

Hence, the variance–covariance matrix of $\tilde{\mathbf{e}}_{t+h}(x)$ can be written as

$$\text{VAR}[\tilde{\mathbf{e}}_{t+h}(x)|\mathbf{y}_1(x), \mathbf{y}_2(x), \dots, \mathbf{y}_t(x)] = \mathbf{S}\mathbf{P}\mathbf{W}_h(x)\mathbf{P}'\mathbf{S}, \quad (21)$$

where $\mathbf{W}_h(x) = \mathbb{E}[\hat{\mathbf{e}}_{t+h}(x)\hat{\mathbf{e}}_{t+h}'(x)|\mathbf{y}_1(x), \mathbf{y}_2(x), \dots, \mathbf{y}_t(x)]$. By minimizing the trace of the matrix in Eq. (21), we obtain the reconciliation matrix $\mathbf{P}$ given by

$$\mathbf{P} = (\mathbf{S}'\mathbf{W}_h(x)^{-1}\mathbf{S})^{-1}\mathbf{S}'\mathbf{W}_h(x)^{-1}. \quad (22)$$

It is worth noting that the only difference between the approach by Hyndman et al. (2011) and Wickramasuriya et al. (2018) is the variance–covariance matrix used in the estimation of $\mathbf{P}$. Since $\Sigma_h(x)$ is generally neither known nor identifiable, the MinT approach provides a practical solution to the problem. Wickramasuriya et al. (2018) have also given a theoretical justification for the usage of $\mathbf{W}_h(x)$ to replace $\Sigma_h(x)$. In addition, as shown by Wickramasuriya et al. (2018), the MinT reconciliation approach produces reconciled forecasts that are at least as good as the base forecasts and can substantially improve forecast accuracy. Therefore, in this study we adopt the MinT reconciliation approach to reconcile the mortality forecasts at all levels.

⁹ For a detailed proof, please refer to Wickramasuriya et al. (2018), Appendix A.1.

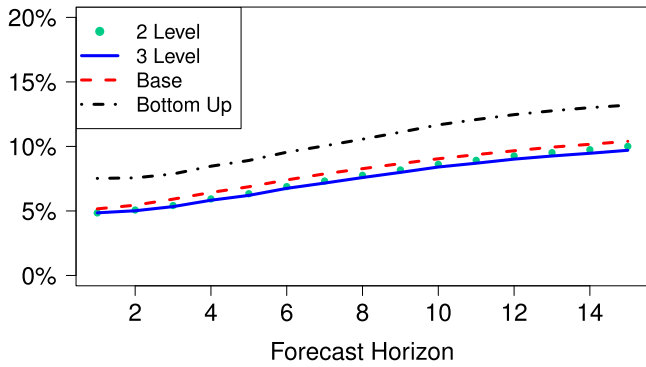


Fig. 6. MAPE over 1–15 year forecast horizons for the all-cause mortality, based on: 2-level MinT reconciliation forecasts (circle), 3-level MinT reconciliation forecasts (solid line), base forecasts (dashed line), and bottom-up forecasts (dotted-dashed line).

3.3. Reconciliation under cause-elimination

Cause-elimination analysis addresses the hypothetical question of what the total mortality experience would be if a particular cause of death were eliminated (Arias et al., 2013). It is an important tool to evaluate the potential impact of efforts devoted to reduce or eliminate mortality risks from certain causes, and has attracted wide attention in different fields (see e.g. Keyfitz, 1977; Manton et al., 1976; Alai et al., 2015, 2018). In this section we illustrate how to conduct this analysis using the reconciliation methodology introduced in Section 3.2.

We focus on the elimination of one single cause j_i . The reconciliation equation becomes

$$\hat{\mathbf{y}}_{T+h}(x) = \mathbf{S}_{j_i} \mathbf{P}_{j_i} \hat{\mathbf{y}}_{T+h}(x), \quad (23)$$

where $\mathbf{S}_{j_i}$ and $\mathbf{P}_{j_i}$ represent the modified summing and reconciliation matrix respectively. We further define these matrices as

- $\mathbf{S}_{j_i}$ is a $(C + J + 1) \times (J - 1)$ matrix which can be obtained from removing the (j_i) th column in $\mathbf{S}$.
- $\mathbf{P}_{j_i}$ is a $(J - 1) \times (C + J + 1)$ matrix which is calculated as $(\mathbf{S}_{j_i}' \mathbf{W}_h(x)^{-1} \mathbf{S}_{j_i})^{-1} \mathbf{S}_{j_i}' \mathbf{W}_h(x)^{-1}$.

For instance, if cancer cause of death were eliminated in the future, the corresponding summing matrix in Eq. (9) becomes

$$\mathbf{S} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & & & & & & \\ \vdots & & & & & & \\ 0 & & & & & & \mathbf{I}_6 \end{pmatrix}. \quad (24)$$

Note that the 5th row in the matrix simply implies the constraint that the reconciled forecast of cancer mortality is 0.

4. Empirical application

In this section we apply the MinT reconciliation approach described in Section 3 to the U.S. male mortality data. Two hierarchical structures are considered. The first structure is a simple 2-level hierarchy given in Fig. 1, in which all cause-of-death mortality rates directly sum up to the top level, i.e. all-cause mortality rates. The second structure is the 3-level hierarchy given in Fig. 5 based on the cluster analysis in Section 2.3.

We first introduce the mortality model used to generate the base mortality forecasts at each level. Second, we compare the out-of-sample forecast accuracy of the MinT reconciliation approach with that of the base forecasts as well as the bottom-up method. Finally, based on the reconciliation method described in Section 3.3, we evaluate the impact of hypothetical elimination of cancer deaths and external deaths defined in this study on the aggregate mortality rates in 2050.

4.1. The Lee–Carter model

Introduced in the early 1990s, the Lee–Carter model (Lee and Carter, 1992) is one of the earliest stochastic mortality models in the literature. It marks the beginning of a new era of mortality modeling. In this paper, we choose the Lee–Carter to model and forecast total mortality rates as well as each individual cause-specific mortality rates.¹⁰ The Lee–Carter model is specified as follows

$$\log m_t(x) = a(x) + b(x)\kappa_t + \varepsilon_t(x), \quad (25)$$

$$\kappa_t = d + \kappa_{t-1} + \epsilon_t, \quad (26)$$

for $x \in (1, N)$ and $t \in (1, T)$. $a(x)$ measures the average level of mortality for age x . κ_t is a period effect capturing the overall mortality trend and is usually modeled by a random walk process with drift d . $b(x)$ measures the rate of decline in log mortality rate at age x in response to κ_t . $\varepsilon_t(x)$ and ϵ_t are normally distributed i.i.d. error terms.

We fit separate Lee–Carter models to all-cause mortality rates (top-level), cluster-specific mortality rates (middle-level), and cause-specific mortality rates (bottom-level) described in Section 2. Base mortality forecasts at each level can be obtained by extrapolating the corresponding period effect κ_t . The bottom-up and MinT reconciled forecasts are then calculated according to the formulae presented in Sections 3.1 and 3.2.

4.2. Out-of-sample forecast accuracy

To evaluate the out-of-sample forecast performance of the proposed approach, in this section we estimate the Lee–Carter model using mortality data at each level from 1970 to 2000, and project mortality rates from 2001 to 2015. For each forecast horizon, we compare the mean absolute percentage error (MAPE) of base forecasts, the bottom-up forecasts and MinT reconciliation forecasts. Formally, the MAPE over forecast horizon h and age range 1–85 is defined as

$$\text{MAPE} = \frac{1}{85 \times h} \sum_{x=1}^{85} \sum_{u=1}^h \left| \frac{\log \hat{m}_{2000+u}(x) - \log m_{2000+u}(x)}{\log m_{2000+u}(x)} \right| \times 100\%.$$

Table 2 shows the MAPE with $h = 15$ generated by the 2-level and the 3-level MinT reconciled forecasts, the base forecasts, and the bottom-up forecasts. We see that the reconciled forecasts lead to the smallest MAPE, i.e., more accurate forecasts in all cases. Moreover, the 3-level reconciled forecasts are more accurate than the 2-level reconciled forecasts in all cases except for mental and vascular.

Fig. 6 displays the MAPE over forecast horizon 1 to 15 years for the all-cause mortality rate. First, we see that the bottom-up forecasts, which ignore the dependence structure among causes of death, lead to the poorest forecasting accuracy. Secondly, the base forecasts generally produce better forecasting accuracy than the bottom-up forecasts, while still under-performing relative

¹⁰ We have also employed the Hyndman and Ullah (2007) model to produce base mortality forecasts, and arrived at the same conclusions with those based on the Lee–Carter model.

Table 2

MAPE for $h = 15$ by the 2-level and the 3-level reconciled forecasts, the base forecasts, and the bottom-up forecasts.

	2-level MinT	3-level MinT	Base	Bottom-up
Total	10.00%	9.70%	10.40%	13.20%
Cancer	17.98%	16.96%	20.06%	–
Diabetes	23.51%	21.78%	25.48%	–
External	19.11%	19.07%	20.62%	–
Influenza	37.17%	35.62%	38.36%	–
Mental	19.65%	20.02%	19.96%	–
Nephritis	59.60%	59.31%	61.31%	–
Vascular	16.37%	16.70%	17.37%	–
Other	26.64%	25.22%	34.46%	–

to MinT reconciliation forecasts. Overall, the MinT reconciliation approach gives the best forecast performance over all forecast horizons considered in this study. More importantly, further improvement in forecast accuracy is achieved by implementing the 3-level hierarchical structure suggested by the cluster analysis in Section 2.3. It is therefore evident that by combining cause-specific mortality data with similar profiles in age and time dimensions, we achieve improvements in forecast accuracy across all levels.

We also produce similar MAPE plots for each individual cause-of-death in Fig. 7. It can be seen that the reconciled forecasts lead to lower MAPE than the base forecasts in most scenarios with the only exceptions being *mental* disease, where both methods have very close values in MAPE. Generally, the 3-level reconciled forecasts are at least as accurate as 2-level reconciled forecasts, except for *mental* causes and *vascular* diseases, where the 3-level reconciled forecasts lead to slightly higher MAPE values than the 2-level reconciled forecasts. If the objective is only to forecast mortality from mental and vascular causes, then on the basis of the evidence presented here (and applying the usual caveats concerning a limited sample) we would recommend the use of the 2-level hierarchy. In all other cases including the case where a single set of coherent forecasts is desired for the full hierarchy, the 3-level hierarchy should be preferred.

We also performed the Diebold–Mariano test (Diebold and Mariano, 2002) on the MAPEs from the 3-level MinT approach and the base forecasts for both the total-level mortality rates and each cause-specific mortality rate with forecast horizon 1–15 years. Our test results show that for all cases where the 3-level MinT approach outperforms the base forecasts, the null hypothesis is rejected at 1% level of significance. In summary, we see that reconciliation is indeed a powerful tool to improve forecast accuracy for cause-specific mortality rates, and again the clustering-based construction of the 3-level hierarchical structure further enhances the forecast performance of the method.

4.3. Mortality projection in 2050

In this section, we project the all-cause mortality rates in 2050, and illustrate results from base forecasts, the bottom-up method, and the MinT reconciliation method. We focus on MinT reconciliation mortality projections based on the 3-level hierarchical structure shown in Fig. 5.¹¹ Fig. 8 shows the logarithm of projected all-cause age-specific mortality rates in 2050.

Among the three approaches, we can see that the bottom-up approach generally leads to the highest mortality projections while base forecasts are generally the lowest, especially among the younger age groups (below 20) and older age groups (above

50). This observation is consistent with Wilmoth (1995), who suggested that projecting the cause-specific mortality rates separately and then summing them up would lead to more pessimistic projections compared to directly projecting the all-cause mortality rates. Based on our analysis, the projected mortality rate based on the bottom-up method can be more than 30% higher than the base forecasts for some old ages.

We also observe that the reconciled forecasts lie between the base forecasts and bottom-up forecasts. This phenomenon is not difficult to understand. The total mortality rates are the smoothest data series in the hierarchy. As a result of aggregation, there is a loss of information from cause-specific and cluster-specific mortality rates. Consequently, base forecasts of total mortality may underestimate the mortality risks in the future and give the most optimistic forecasts. On the other hand, bottom-up method utilizes all the cause-specific mortality data. However, the method ignores the dependence structure among different causes, which is well recognized in the literature (Alai et al., 2015; Arnold and Sherris, 2015). Therefore, by directly adding up the base forecasts of each cause-of-death, the bottom-up method runs a risk of overestimating future mortality rates.

To overcome these aforementioned limitations, the proposed reconciliation mortality forecasting approach combines information on both the aggregate and disaggregate levels, and taking into account the variance–covariance structures of errors from each data series. As a result, the MinT reconciliation approach achieves more reliable mortality projections into the future.

4.4. Cause elimination

In this section, we apply the cause-elimination reconciliation method described in Section 3.3 to *cancer* and *external*, respectively, and project the age-specific mortality rates until 2050. The motivation for us to focus on these two causes of death is as follows. First, the elimination of cancer deaths will have a considerable impact on the life insurance industry. Currently, the majority of life insurance policies sold in developed countries are under the “preferred” or the “super preferred” insurance rating classifications which are the two lowest rate classes.¹² One of the general requirements for these low rate classes is no cancer history or no family history of cancer deaths. Therefore, the elimination of cancer deaths would potentially change the rating scheme for life insurance products.

Secondly, external causes are one of the most important drivers of mortality amongst young and middle age populations in the U.S. In particular, over 70% of deaths among teenagers aged 12 to 19 years during 1999 to 2006 were due to unintentional injury, homicide, and suicide (Centers for Disease Control and Prevention, 2010). Moreover, mortality related to drug overdose, suicide, and alcohol consumption has been increasing for the middle-age males in the U.S. since the end of the 20th century (Case and Deaton, 2017; Kochanek et al., 2017). In particular, the mortality increase in this age range due to external causes has offset the mortality improvements brought by other causes, and has led to decline in overall life expectancy at birth in the U.S. in 2014 and 2015 (Xu et al., 2016). Therefore, it is also important to evaluate what the all-cause mortality would be if effective regulations were put into effect to substantially reduce mortality risks due to external causes.

In this section, we compare three sets of mortality projections:

1. The “benchmark” all-cause forecasts — the reconciled projections of the total mortality rates when no cause is eliminated.

¹¹ We have also conducted the analysis using MinT reconciliation approach based on the 2-level hierarchical structure and found very similar results. These results are available upon request.

¹² According to the Scor Global Life America Publication *Scor View*, September 2015, about 70% of life insurance policies in the U.S. belonged to these two rating classifications in 2014.

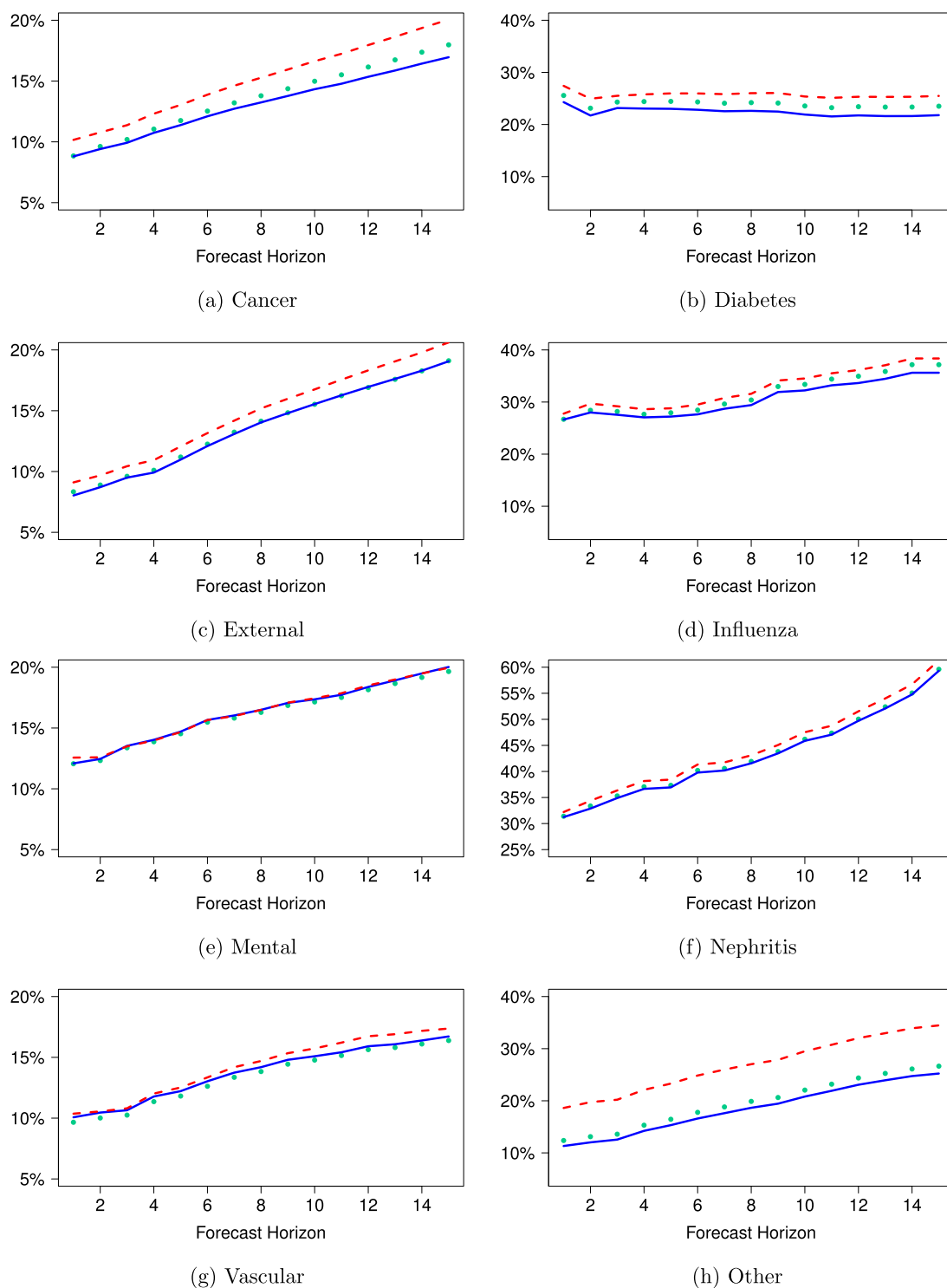


Fig. 7. MAPE over 1–15 year forecast horizons for cause-specific mortality, based on: the 2-level MinT reconciliation forecasts (circle), the 3-level reconciliation forecasts (solid line), and the base forecasts (dashed line).

2. The “reconciled” elimination forecasts — the reconciled projections of the total mortality rates where the cause of interest is eliminated.
3. The “independent” elimination forecasts — the benchmark forecasts minus the base forecasts of the mortality rates of the cause of interest.

Fig. 9 displays the aforementioned three sets of forecasts in 2050, with *cancer* death rates being eliminated in the reconciled forecasts and independent forecasts. First, we see that the

elimination of *cancer* leads to lower projected mortality rates in both the reconciled and the independent elimination forecasts, in particular for ages between 50 and 80. More importantly, the independent forecasts are lower than the reconciled ones for almost all age groups. The intuition behind this is that the reconciliation procedure takes into account the competing nature of different causes of death. Specifically, if the individuals do not die of cancer, they would eventually die of some other causes. As a result, mortality rates of other causes would increase, and

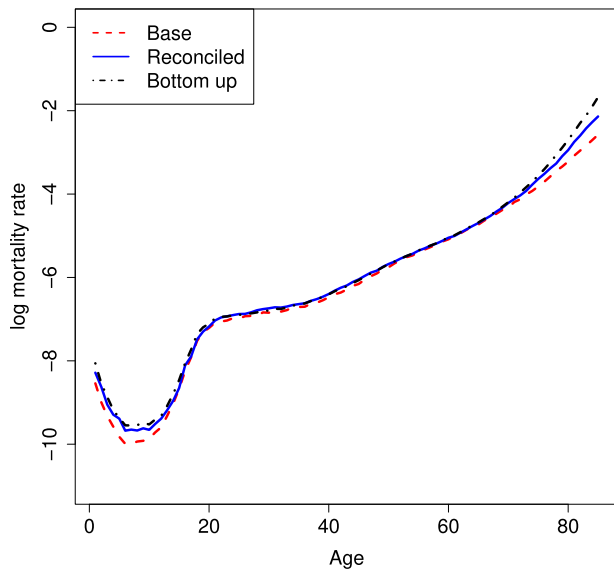


Fig. 8. The logarithm of projected all-cause mortality rates in 2050: base forecasts (dashed line), the 3-level reconciled forecasts (solid line), and the bottom-up forecasts (dotted-dashed line).

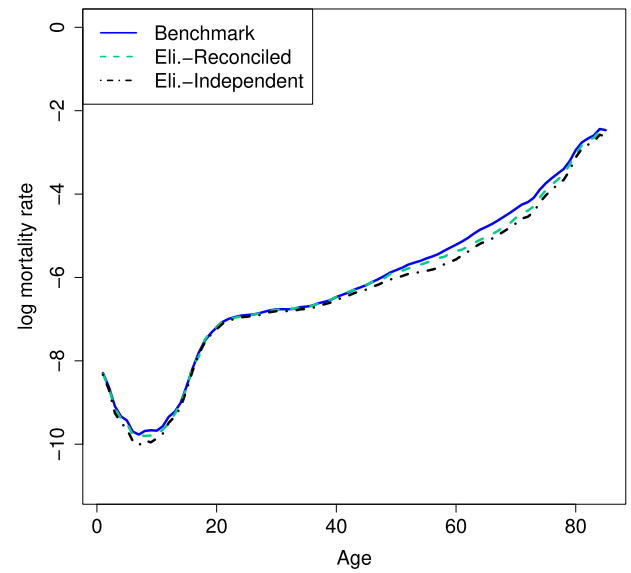


Fig. 9. The logarithm of projected mortality rates in 2050: the benchmark forecasts (dashed line), the reconciled forecasts (solid line), and the independent forecasts (dotted-dashed line) when cancer is eliminated.

thus (partially) offset the mortality reduction brought by the elimination of *cancer*. In contrast, the independent elimination approach assumes that *cancer* is independent of all other causes, and therefore may overestimate the total mortality reduction due to elimination of cancer.

To illustrate the impact of cancer elimination more clearly, in Fig. 10, we show the three sets of mortality forecasts from 2016 to 2050 for a selection of ages. Since the impact of cancer elimination is more apparent for older age groups, we have chosen to show forecasts for ages 55, 60, 65, and 70. In line

with Fig. 9, we see that throughout the 35-year forecast horizon, the reconciled forecasts fall between the benchmark and the independent forecasts for all four age groups.

5. Conclusion

Fig. 11 shows mortality projections in 2050 under the hypothetical scenario where *external* causes are eliminated. In contrast to *cancer*, the elimination of *external* leads to mortality improvements at younger ages, especially from 15 to 50. Also, we see

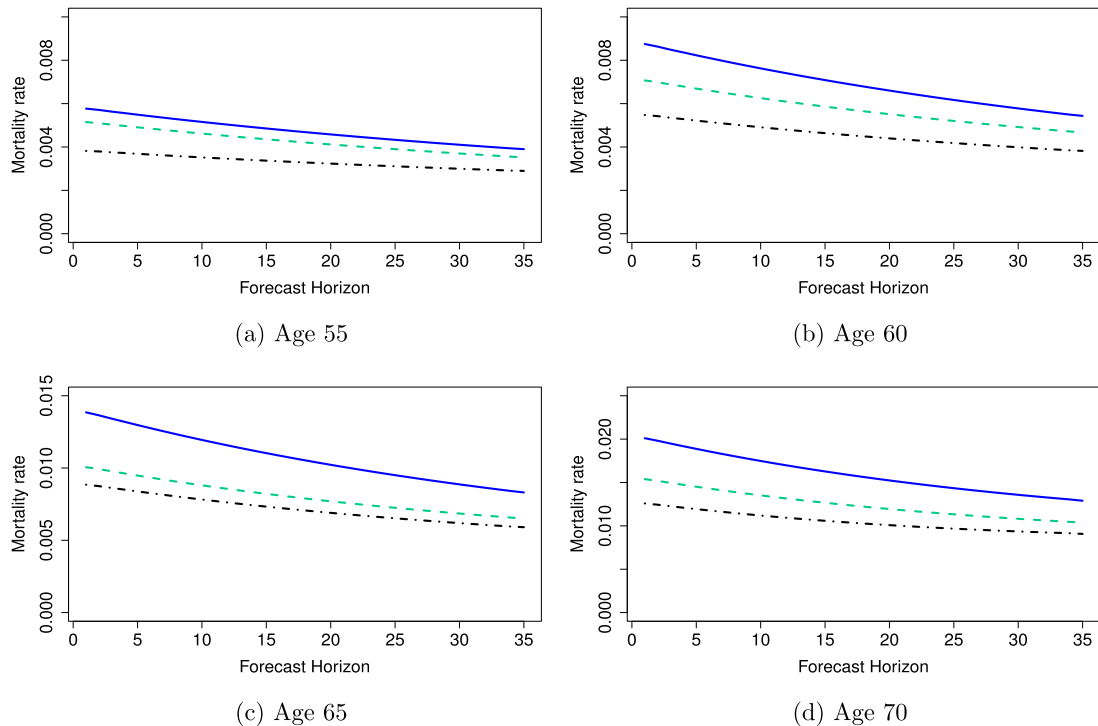


Fig. 10. The projected mortality rates of ages 55, 60, 65, and 70: the benchmark forecast (solid line), the reconciled forecast (dashed line), and the independent forecast (dotted-dashed line) when cancer is eliminated.

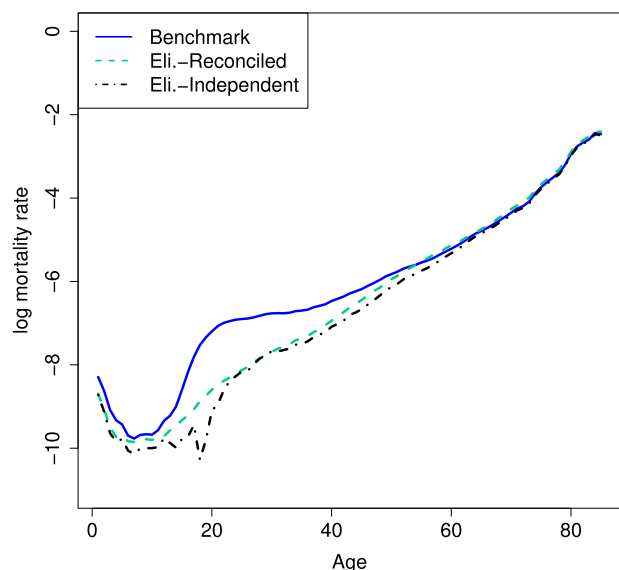


Fig. 11. The logarithm of projected mortality rates in 2050: the benchmark forecasts (dashed line), the reconciled forecasts (solid line), and the independent forecasts (dotted-dashed line) when *external* is eliminated.

that the reconciled forecasts are higher than the independent forecasts for most ages. In particular, the independent forecast predicts nearly 40% mortality reduction at 18, whereas the predicted reduction is less than 20% by the reconciled forecast. Moreover, the reconciled projected mortality rates above 60 are slightly higher than the benchmark forecasts. This observation is reasonable, since if individuals do not die of external causes at young ages, they would eventually die of other causes, possibly chronic diseases, when they become older. As a result, the overall

mortality rates at old ages would increase. Fig. 12 displays the projected mortality rates for ages 10, 20, 30, and 40 from 2015 to 2050. Similarly, we observe that in almost all cases the reconciled forecasts fall between the benchmark and the independent forecasts for all forecast horizons.

In this paper, we propose a coherent forecasting approach to cause-of-death mortality modeling via trace minimization reconciliation. The forecast reconciliation approach allows us to utilize mortality experiences at all levels in the forecasting procedure, and ensure coherence across forecasts is achieved. A cluster analysis is also carried out to create additional levels in the hierarchy of mortality rates.

We apply the proposed approach to U.S. male mortality data from 1970 to 2015. The out-of-sample forecasting results show that our approach gives superior performance for both cause-specific and aggregate mortality projections, compared to the base forecasts and the bottom-up forecasts. Moreover, by grouping causes with similar mortality patterns, forecasting accuracy can be further improved. The forecast reconciliation approach is also applied to a scenario-based analysis where cancer and external causes are eliminated. We find that the forecast reconciliation approach is able to take into account the competing nature of different causes of death, and thus leads to more sensible projected mortality rates than the case where the eliminated cause of death is assumed to be independent of other causes.

An interesting extension to our study would be to use the MinT approach for the reconciliation of forecast intervals or densities. This is a growing area of research in general. For instance, Shang (2017) produces forecast intervals, using weighted least squares reconciliation rather than MinT, while Gamakumara et al. (2018) and Jeon et al. (2018) propose methods to produce an entire reconciled forecast density. Probabilistic forecast reconciliation for cause-of-death mortality rates promises to be an important avenue for future research.

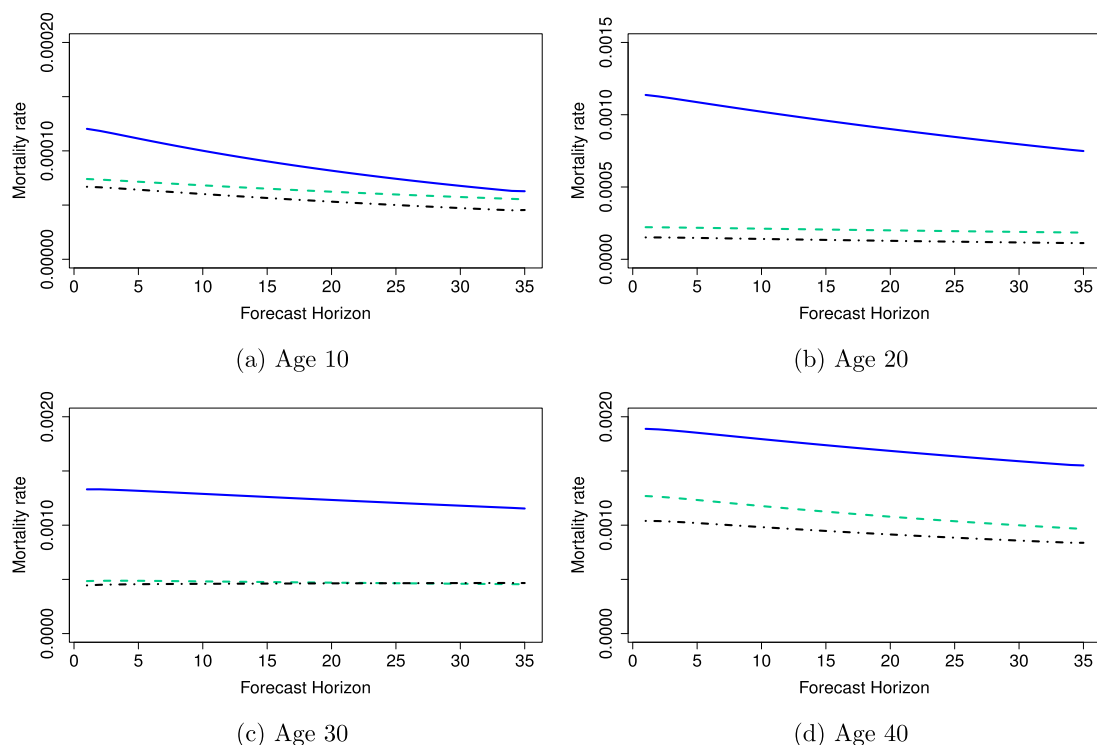


Fig. 12. The projected mortality rate of ages 10, 20, 30, and 40 by the benchmark forecast (solid line), the reconciled forecast (dashed line), and the independent forecast (dotted-dashed line) when *external* is eliminated.

References

- Alai, D.H., Arnold, S., Bajekal, M., Villegas, A.M., 2018. Mind the gap: a study of cause-specific mortality by socioeconomic circumstances. *N. Am. Actuar. J.* 22 (2), 161–181.
- Alai, D.H., Arnold, S., Sherris, M., 2015. Modelling cause-of-death mortality and the impact of cause-elimination. *Ann. Actuar. Sci.* 9 (1), 167–186.
- Arias, E., Heron, M., Tejada-Vera, B., et al., 2013. National Vital Statistics Reports. National Center for Health Statistics.
- Arnold, S., Sherris, M., 2013. Forecasting mortality trends allowing for cause-of-death mortality dependence. *N. Am. Actuar. J.* 17 (4), 273–282.
- Arnold, S., Sherris, M., 2015. Causes-of-death mortality: what do we know on their dependence? *N. Am. Actuar. J.* 19 (2), 116–128.
- Arnold, S., Sherris, M., 2016. International cause-specific mortality rates: new insights from a cointegration analysis. *Astin Bull.* 46 (1), 9–38.
- Athanasopoulos, G., Ahmed, R.A., Hyndman, R.J., 2009. Hierarchical forecasts for Australian domestic tourism. *Int. J. Forecast.* 25 (1), 146–166.
- Booth, H., Tickle, L., 2008. Mortality modelling and forecasting: a review of methods. *Ann. Actuar. Sci.* 3 (1–2), 3–43.
- Borges, C.E., Penya, Y.K., Fernandez, I., 2013. Evaluating combined load forecasting in large power systems and smart grids. *IEEE Trans. Ind. Inf.* 9 (3), 1570–1577.
- Capistrán, C., Constandse, C., Ramos-Francia, M., 2010. Multi-horizon inflation forecasts using disaggregated data. *Econ. Model.* 27 (3), 666–677.
- Case, A., Deaton, A., 2017. Mortality and morbidity in the 21st century. In: *Brookings Papers on Economic Activity*, Vol. 2017. NIH Public Access, p. 397.
- Caselli, G., Vallin, J., Marsili, M., 2006. How useful are the causes of death when extrapolating mortality trends. an update. In: *Social Insurance Studies from the Swedish Social Insurance*, Vol. 4. pp. 9–36.
- Centers for Disease Control and Prevention (2010) Mortality among teenagers aged 12–19 years: United states, 1999–2006. Retrieved April, 28:2011.
- Dangerfield, B.J., Morris, J.S., 1992. Top-down or bottom-up: aggregate versus disaggregate extrapolations. *Int. J. Forecast.* 8 (2), 233–241.
- Diebold, F.X., Mariano, R.S., 2002. Comparing predictive accuracy. *J. Bus. Econom. Statist.* 20 (1), 134–144.
- Dimitrova, D., Haberman, S., Kaishev, V., 2013. Dependent competing risks: cause elimination and its impact on survival. *Insurance Math. Econom.* 53 (2), 464–477.
- Escarela, G., Carriere, J.F., 2003. Fitting competing risks with an assumed copula. *Stat. Methods Med. Res.* 12 (4), 333–349.
- Frees, E.W., Valdez, E.A., 1998. Understanding relationships using copulas. *N. Am. Actuar. J.* 2 (1), 1–25.
- Gamakumara, P., Panagiotelis, A., Athanasopoulos, G., Hyndman, R.J., et al., 2018. Probabilistic forecasts in hierarchical time series. In: *Technical Report*. Monash University, Department of Econometrics and Business Statistics.
- Giordano, G., Haberman, S., Russolillo, M., 2018. Three-way data analysis applied to cause specific mortality trends. In: *Demography and Health Issues*. Springer, pp. 121–130.
- Graybill, F.A., 1983. *Matrices with applications in statistics*.
- Gross, C.W., Sohl, J.E., 1990. Disaggregation methods to expedite product line forecasting. *J. Forecast.* 9 (3), 233–254.
- Hyndman, R.J., Ahmed, R.A., Athanasopoulos, G., Shang, H.L., 2011. Optimal combination forecasts for hierarchical time series. *Comput. Statist. Data Anal.* 55 (9), 2579–2589.
- Hyndman, R.J., Athanasopoulos, G., 2018. *Forecasting: Principles and Practice*. OTexts.
- Hyndman, R.J., Athanasopoulos, G., et al., 2014. Optimally reconciling forecasts in a hierarchy. *Foresight: Int. J. Appl. Forecast.* (35), 42–48.
- Hyndman, R.J., Ullah, M.S., 2007. Robust forecasting of mortality and fertility rates: a functional data approach. *Comput. Statist. Data Anal.* 51 (10), 4942–4956.
- Jeon, J., Panagiotelis, A., Petropoulos, F., 2018. Reconciliation of probabilistic forecasts with an application to wind power, <https://arxiv.org/abs/1808.02635>.
- Kahn, K.B., 1998. Revisiting top-down versus bottom-up forecasting. *J. Bus. Forecast.* 17 (2), 14.
- Kaishev, V., Dimitrova, D., Haberman, S., 2007. Modelling the joint distribution of competing risks survival times using copula functions. *Insurance Math. Econom.* 41 (3), 339–361.
- Keyfitz, N., 1977. What difference would it make if cancer were eradicated? an examination of the taeuber paradox. *Demography* 14 (4), 411–418.
- Kochanek, K.D., Arias, E., Bastian, B.A., for Health Statistics, N.C., et al., 2017. The effect of changes in selected age-specific causes of death on non-hispanic white life expectancy between 2000 and 2014. In: *Heart Disease*, Vol. 1. pp. 0–581.
- Lee, R.D., Carter, L.R., 1992. Modeling and forecasting US mortality. *J. Amer. Statist. Assoc.* 87 (419), 659–671.
- Li, H., Lu, Y., 2018. Modeling cause-of-death mortality using hierarchical archimedean copula. *Scand. Actuar. J.* 1–26.
- Manton, K.G., Stallard, E., Vaupel, J.W., 1986. Alternative models for the heterogeneity of mortality risks among the aged. *J. Amer. Statist. Assoc.* 81 (395), 635–644.
- Manton, K.G., Tolley, H.D., Poss, S.S., 1976. Life table techniques for multiple-cause mortality. *Demography* 13 (4), 541–564.
- McNown, R., Rogers, A., 1992. Forecasting cause-specific mortality using time series methods. *Int. J. Forecast.* 8 (3), 413–432.
- Michael, J., Krause, J.S., Lammertse, D.P., 1999. Recent trends in mortality and causes of death among persons with spinal cord injury. *Arch. Phys. Med. Rehabil.* 80 (11), 1411–1419.
- Murtagh, F., Legendre, P., 2014. Wards hierarchical agglomerative clustering method: which algorithms implement wards criterion? *J. Classification* 31 (3), 274–295.
- Schwarzkopf, A.B., Tersine, R.J., Morris, J.S., 1988. Top-down versus bottom-up forecasting strategies. *Int. J. Prod. Res.* 26 (11), 1833–1843.
- Shang, H.L., 2017. Reconciling forecasts of infant mortality rates at national and sub-national levels: grouped time-series methods. *Popul. Res. Policy Rev.* 36 (1), 55–84.
- Shang, H.L., Haberman, S., 2017. Grouped multivariate and functional time series forecasting: an application to annuity pricing. *Insurance Math. Econom.* 75, 166–179.
- Shang, H.L., Hyndman, R.J., 2017. Grouped functional time series forecasting: an application to age-specific mortality rates. *J. Comput. Graph. Statist.* 26 (2), 330–343.
- Shlifer, E., Wolff, R., 1979. Aggregation and proration in forecasting. *Manage. Sci.* 25 (6), 594–603.
- Syntetos, A.A., Babai, Z., Boylan, J.E., Kolassa, S., Nikolopoulos, K., 2016. Supply chain forecasting: theory, practice, their gap and the future. *European J. Oper. Res.* 252 (1), 1–26.
- Tabeau, E., Ekamper, P., Huisman, C., Bosch, A., 1999. Improving overall mortality forecasts by analysing cause-of-death, period and cohort effects in trends. *Eur. J. Population/Revue eur. Démogr.* 15 (2), 153–183.
- Ward, Jr., J.H., 1963. Hierarchical grouping to optimize an objective function. *J. Amer. Statist. Assoc.* 58 (301), 236–244.
- Weale, M., 1988. The reconciliation of values, volumes and prices in the national accounts. *J. Roy. Statist. Soc. Ser. A* 151 (1), 211–221.
- Wickramasuriya, S.L., Athanasopoulos, G., Hyndman, R.J., 2018. Optimal forecast reconciliation for hierarchical and grouped time series through trace minimization. *J. Am. Stat. Assoc.* 1–16. <http://dx.doi.org/10.1080/01621459.2018.1448825>.
- Wilmoth, J.R., 1995. Are mortality projections always more pessimistic when disaggregated by cause of death? *Math. Popul. Stud.* 5 (4), 293–319.
- Xu, J., Murphy, S.L., Kochanek, K.D., Arias, E., 2016. Mortality in the United States, 2016. In: *NCHS Data Brief*, Vol. 293. pp. 1–8.
- Zellner, A., Tobias, J., 2000. A note on aggregation, disaggregation and forecasting performance. *J. Forecast.* 19 (5), 457–465.